

Cover memorandum for patent counsel

To: Wood Herron & Evans LLP — patent counsel (to be assigned; referral via Lori Krafte)
From: Todd Wichmann, Instar Brands LLC (Loveland, Ohio)
Date: July 20, 2026
Re: Review draft — utility patent application, instrument-based promotional settlement system ("SACM / Meme Token Credit" project, Aladdin's Castle program)

This is an inventor-prepared review draft, not a filing. It is organized as a complete application (specification, four informal figures, twenty claims, abstract) so your team can evaluate scope quickly and convert it efficiently. Notes for your review:

1. **Inventor / assignee.** Sole inventor: Todd Wichmann. Intended assignee: Instar Brands LLC, an Ohio limited liability company. Please advise on assignment and any micro/small-entity status.
2. **Filing strategy.** The inventor's preference is to file a **provisional promptly** to establish priority before any public launch or marketing of the mechanism, with the non-provisional to follow within the year. No public disclosure of the mechanism has been made to date; the commercial app design exists only in private materials.
3. **Claims ladder.** Independent claims 1 (method), 10 (system), and 20 (medium) are drafted broadly; dependent claims are deliberate fallback rungs (locked membership asset, dual-rail anchoring, offline one-time codes, store-and-forward terminals, no-reversal-path, physical dispensing, event taxonomy, aggregate measurement). The specification is written to support amendment down each rung. Please calibrate the top rungs against your search results.
4. **Prior-art references.** The References section identifies the prior-art families the inventor is aware of **by assignee and subject, without patent numbers**, to avoid propagating unverified citations. A professional search is requested; the distinguishing arguments in the Background and Summary anticipate the expected families (balance-based blockchain loyalty; centralized single-use coupon codes; payment-token systems).
5. **Section 101 posture.** The specification is drafted to emphasize concrete technical improvements (offline-capable redemption terminals, irreversible consensus-verified state transitions, absence of reversal infrastructure, physical dispensing hardware) and to keep motivating theory to background prose. Please prune further if your judgment differs.
6. **Trademark coordination.** "SACM," "Meme Token Credit," "MTC," and "Aladdin's Castle" appear as defined commercial names of embodiments only. Trademark matters are with Lori Krafte separately.
7. **Figures.** FIGS. 1–4 are informal drawings prepared by the inventor; formal drawings to be prepared at filing.

Timing note. The commercial program contemplates a public app launch and marketing of a "\$1 founder's donation" mechanism. Counsel is asked to confirm the priority filing is on file **before** any public disclosure, App Store listing, or partner distribution beyond NDA.

SYSTEM AND METHOD FOR INSTRUMENT-BASED SETTLEMENT OF PROMOTIONAL CREDITS AND MARKETING EVENTS USING UNIQUE SINGLE-USE TOKENS WITH IRREVERSIBLE COMMITMENT ON A DISTRIBUTED LEDGER

CROSS-REFERENCE TO RELATED APPLICATIONS

[0001] Not applicable.

FIELD

[0002] This disclosure relates to electronic settlement of promotional instruments, and more particularly to systems and methods in which each promotional credit or marketing event is bound one-to-one to a unique settlement instrument whose redemption state is verified and irreversibly committed at a point of settlement, including by commitment to a distributed ledger.

BACKGROUND

[0003] Promotional instruments — coupons, rebates, product-sample offers, and loyalty rewards — move very large sums between manufacturers, retailers, and consumers. In prevailing practice these instruments are settled through centralized clearing: a redemption occurs at a point of sale, evidence of the redemption is transported to a clearinghouse, and reconciliation among the funding manufacturer, the honoring retailer, and intermediaries completes days or weeks later. During that interval each redeemed instrument exists in an indeterminate settlement state: it may prove valid, duplicated, counterfeit, misreported, or disputed.

[0004] This settlement latency has two well-documented costs. The first is fraud. Industry analyses estimate direct coupon fraud in the United States at several hundred million dollars annually, with broader promotion-abuse losses estimated in the tens of billions of dollars annually; individual counterfeiting operations have produced tens of millions of dollars in losses. A significant fraction of merchants report knowingly tolerating a level of promotional abuse because the cost of detection under existing infrastructure exceeds the recovery. The dominant fraud modes — duplicate redemption of a single instrument and redemption of counterfeit instruments — are failures of settlement integrity: the same promise is honored more than once, or a promise that was never issued is honored.

[0005] The second cost is measurement loss, commonly called breakage. Substantial percentages of issued loyalty value are never redeemed or expire unobserved, with outstanding unredeemed value estimated in the hundreds of billions of dollars in the United States alone. Under centralized clearing the issuer often cannot distinguish, at the level of an individual instrument and in useful time, among redemption,

expiration, duplication, and loss. Promotional spending is committed against outcomes that are never individually measured.

[0006] Both costs share a root cause: existing infrastructure settles promotional instruments as entries in mutable, balance-based records reconciled after the fact, rather than as discrete instruments whose terminal state is committed once, verifiably, at the moment of settlement. A further consequence is under-issuance: because every outstanding instrument carries settlement uncertainty, issuers rationally throttle promotional volume below what the market would otherwise support.

[0007] Distributed-ledger loyalty systems have been proposed, and are believed to be the subject of prior patent filings by financial-services and technology companies. Known approaches are predominantly *balance-based*: loyalty points are recorded as fungible balances on a ledger, transferred among accounts, and reconciled in aggregate. Such systems relocate the account book but retain its structure; they do not bind an individual promise to an individual instrument, and therefore do not make the settlement of any single promise individually verifiable or individually final. Separately, centralized single-use digital coupon codes are known; these provide once-only redemption within a single issuer's database but require every settling party to trust that issuer's private records, provide no independently verifiable commitment among manufacturers, retailers, and clearing parties, and retain the reversal and dispute machinery of the centralized model.

[0008] Payment-oriented token systems illustrate a further mismatch. In payment systems, irreversibility is a liability to be mitigated: a payer has surrendered value and may require recourse, so chargeback and refund infrastructure is layered on. A promotional instrument differs categorically: the consumer surrenders nothing to receive it. There is no payer requiring protection, and reversal infrastructure serves no protective purpose while creating the very pathway that duplicate-redemption fraud exploits. Applicant is not aware of settlement systems designed around this property of the promotional instrument class.

[0009] There is accordingly a need for promotional settlement infrastructure in which each promise is bound to exactly one uniquely identifiable instrument; in which redemption is validated and irreversibly committed at the point of settlement, in a manner verifiable by parties other than the issuer; which functions where consumer devices and merchant terminals lack reliable connectivity; which by design provides no reversal pathway; and which, as a byproduct of settlement, yields a complete, event-level, privacy-preserving record of promotional activity suitable for population-scale measurement and automatic inter-party reconciliation.

SUMMARY

[0010] Aspects of the disclosure provide a settlement system in which a promotional credit is issued bound one-to-one to a unique settlement instrument, is non-transferable and time-limited, and is extinguished exactly once by an irreversible, independently verifiable state transition committed at a point of settlement.

In disclosed embodiments the unique instrument is, or is anchored to, a token recorded on a distributed ledger.

[0011] In one aspect, eligibility to receive credits is gated by verifying, through a read-only query, that a consumer account holds at least a predetermined quantity of a membership asset recorded on a distributed ledger. The membership asset is never spent, transferred, or consumed by operation of the system, and in some embodiments is held in a time-locked vault that prevents its transfer for a period of years, such that the membership asset functions solely as a durable eligibility credential and not as a medium of exchange within the system.

[0012] In another aspect, redemption is validated using a single-use code generated on the consumer's device by a time-based one-time algorithm from a locally stored secret, requiring no network connectivity on the consumer's side at the moment of redemption. A redemption terminal validates the code and causes an atomic state transition that extinguishes the credit; when the terminal itself lacks connectivity, it appends a signed commitment to a store-and-forward queue and synchronizes when connectivity returns, so that settlement proceeds at the point of sale regardless of network state.

[0013] In another aspect, the system deliberately provides no transaction type that reverses a committed redemption. Because the instrument class has no payer requiring recourse, the absence of reversal infrastructure removes both the attack surface exploited by duplicate redemption and the reconciliation and dispute overhead of centralized clearing. Settlement finality is thereby made immediate, which permits issuers to increase promotional volume and velocity without increasing settlement risk.

[0014] In another aspect, committed events — including issuance, redemption, expiration, reimbursement, funding, and verification-attestation events — form an auditable event-level record from which population-scale measurements of promotional activity (including redemption velocity, per-offer consumption, and breakage) are derived in substantially real time, and from which reimbursement obligations between funding parties and honoring parties are computed automatically from the committed events themselves, without a separate clearing intermediary. Impression-type events, where used, are committed as batched pseudonymous digests, and identity verification is committed as a boolean attestation, so that no personally identifying information is written to any ledger.

[0015] In another aspect, commitment of a redemption causes a physical actuation, such as dispensing of physical articles — in a disclosed embodiment, machine-usable amusement tokens accepted only by the operator's machines and recirculated through them — coupling the settlement system to physical apparatus at the point of redemption.

[0016] Embodiments include a dual-rail architecture in which credits are recorded on a private ledger of the settlement system for speed and cost, while state transitions are anchored to a public distributed ledger, individually or as batched cryptographic digests, preserving independent verifiability; and a single-rail

architecture in which each credit is itself a non-transferable token on the distributed ledger, extinguished by a burn transaction. The disclosure encompasses both.

BRIEF DESCRIPTION OF THE DRAWINGS

[0017] FIG. 1 is a block diagram of a promotional settlement system according to embodiments.

[0018] FIG. 2 is a state diagram of the lifecycle of a promotional credit according to embodiments.

[0019] FIG. 3 is a sequence diagram of redemption at a point of settlement, including offline operation, according to embodiments.

[0020] FIG. 4 is a block diagram of event-level recording and aggregate measurement according to embodiments.

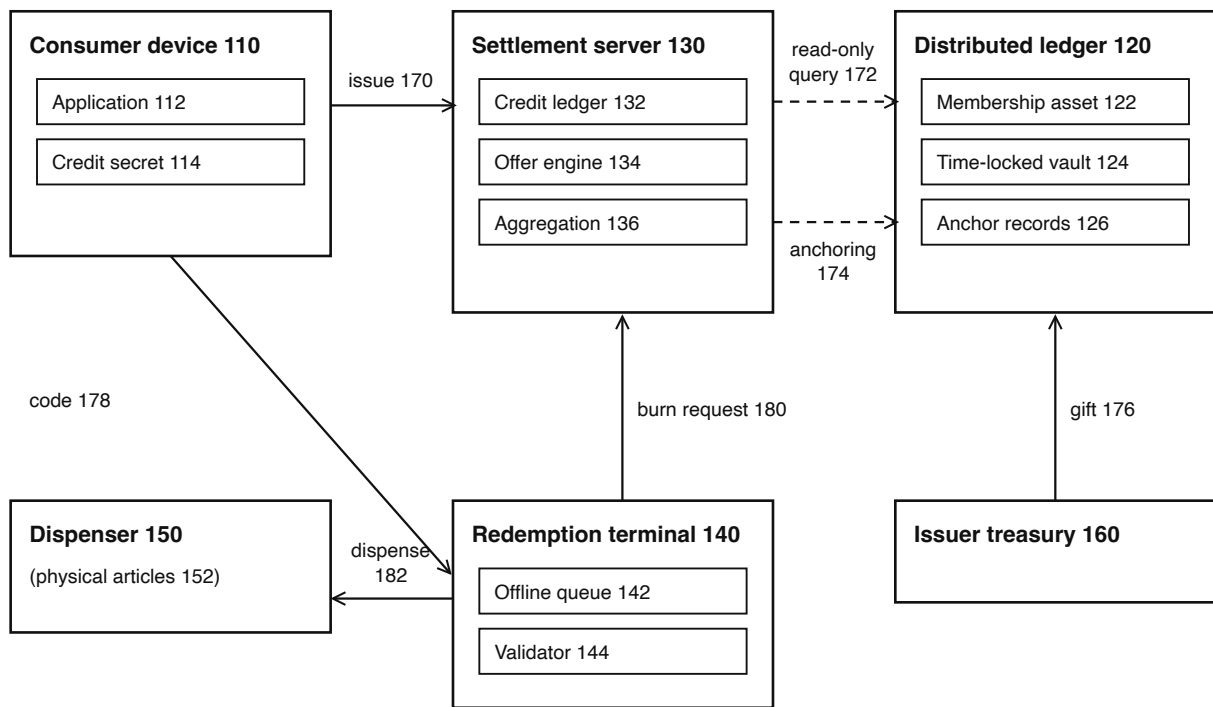


FIG. 1

All transitions one-way; no reversal transaction type exists (see [0035]).

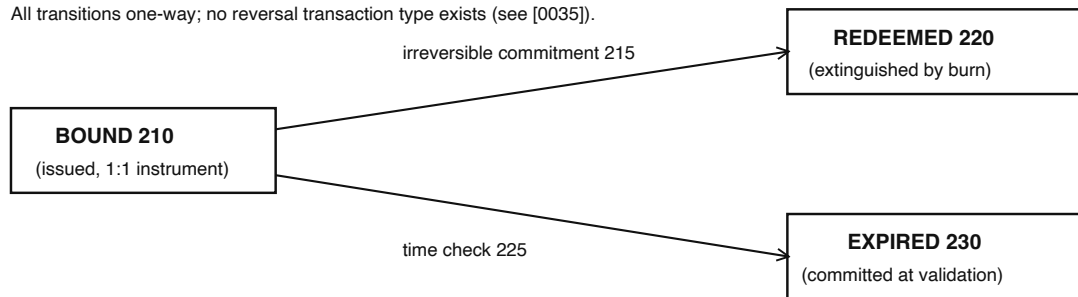


FIG. 2

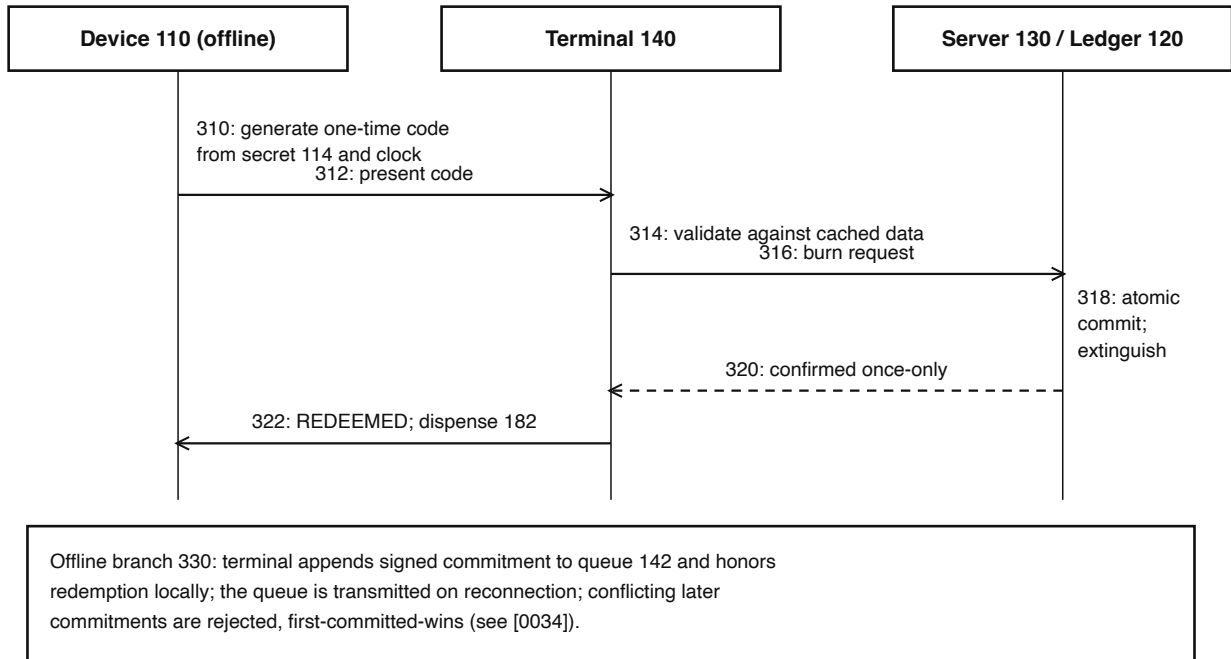


FIG. 3

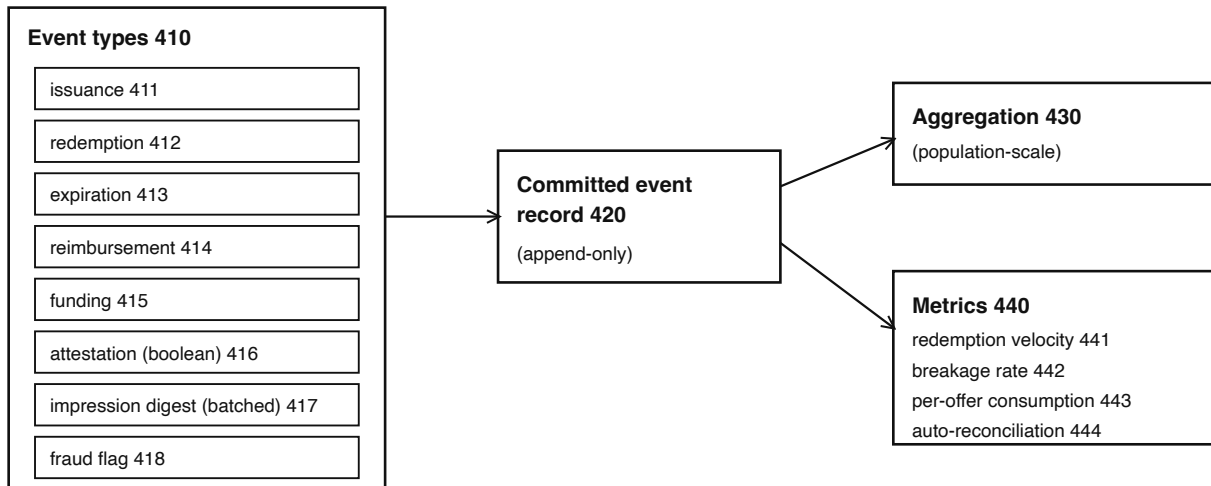


FIG. 4

DETAILED DESCRIPTION

Definitions

[0021] As used herein, a "distributed ledger" is a data store whose state is replicated across a plurality of computing nodes and whose state transitions are validated by a consensus protocol such that no single party can unilaterally alter committed state. Public permissionless ledgers (for example, the Solana ledger) and permissioned multi-party ledgers are both encompassed. A "token" is a uniquely identifiable unit of state recorded on such a ledger. A "burn" is a state transition that permanently renders a token unusable for further transfer or redemption.

[0022] A "membership asset" is a quantity of a ledger-recorded asset whose holding, above a threshold, evidences eligibility, and which is not spent, transferred, or consumed by any operation of the settlement system. In one commercial embodiment the membership asset is a token denominated "\$ACM."

[0023] A "promotional credit" is a non-transferable, time-limited instrument entitling a holder to a defined benefit (a discount, a product sample, machine credit, access, or the like) upon exactly one redemption. In one commercial embodiment a promotional credit is denominated a "Meme Token Credit" or "MTC." A "marketing event" is any discrete, individually recordable occurrence in the promotional lifecycle, including at least issuance, redemption, and expiration of a credit, and further including reimbursement, funding, verification-attestation, impression-digest, and fraud-flag events. A "point of settlement" is the place and moment at which a credit is honored, whether at a physical terminal or a network endpoint.

System overview (FIG. 1)

[0024] Referring to FIG. 1, a settlement system comprises a consumer device 110 executing an application 112 and holding a credit secret 114; a settlement server 130 comprising a credit ledger 132, an offer engine 134, and an aggregation module 136; a distributed ledger 120 recording a membership asset 122, a time-locked vault 124, and anchor records 126; a redemption terminal 140 comprising an offline queue 142 and a validator 144; a dispenser 150 of physical articles 152; and an issuer treasury 160. The settlement server 130 communicates with the distributed ledger 120 through a read-only query interface 172 for eligibility verification and, in dual-rail embodiments, through an anchoring interface 174 for commitment of state-transition digests.

[0025] In embodiments the terminal 140 is a commodity mobile device (for example, a repurposed smartphone) operating in a locked kiosk mode, reading optically displayed codes by camera; near-field-communication reading is an alternative presentation channel. Site infrastructure may include a satellite or cellular uplink with battery backup; however, as described below, no component of the redemption path requires connectivity at the moment of redemption.

Membership gating

[0026] The offer engine 134 issues credits only to accounts for which the read-only query 172 confirms holding of at least a predetermined quantity — in embodiments, a single indivisible unit — of the membership asset 122. Ownership of the queried address is established by a cryptographic signature challenge completed on the device 110, or, for custodially held assets, by delegated authorization to the custodian. The query is the system's only interaction with ledger-held consumer state: the system writes no consumer personal data to any ledger, and the membership asset is never debited, staked, escrowed, or transferred by any system operation.

[0027] In a disclosed embodiment, a consumer without a pre-existing holding acquires eligibility as follows: the consumer purchases a digital item (for example, a commemorative badge) through the host application platform's native purchase mechanism; in response, the issuer treasury 160 transfers a unit of the membership asset from treasury holdings into a time-locked vault 124 associated with the consumer's account, the vault preventing any transfer, sale, or withdrawal of the unit for a predetermined period of years (for example, ten to thirty years). The locked unit thereby functions purely as a durable eligibility credential. The consumer at no point purchases the ledger asset itself, and no fiat-to-asset conversion is performed on the consumer's behalf.

[0028] The separation of the never-consumed membership asset from the consumable credit is deliberate: eligibility is durable and non-monetary, while spendable value exists only as non-transferable, expiring credits. This separation permits the system to confer continuing benefits on holders without the membership asset ever functioning as a medium of exchange within the system.

Issuance and one-to-one binding

[0029] Upon issuance, the offer engine 134 creates a credit record comprising at least a unique credit identifier, an offer definition, an issuance timestamp, an expiration time, an association to the verified account, and a state field initialized to BOUND 210 (FIG. 2). Each credit is bound one-to-one to a unique settlement instrument. In a single-rail embodiment the instrument is a non-transferable token minted on the distributed ledger 120 (for example, using a token standard supporting non-transferability), and extinguishment is a burn transaction of that token. In a dual-rail embodiment the credit record on credit ledger 132 is itself the instrument, and its committed state transitions are anchored to the distributed ledger through interface 174, individually or as a batched cryptographic digest of a plurality of transitions, such that any party may verify after the fact that a claimed transition was committed and that no committed transition has been altered. Both embodiments provide independent verifiability; they differ in per-event cost and latency, and an operator may employ both simultaneously for different offer classes.

[0030] Credits are non-transferable by construction: no operation of the system transfers a credit between accounts. In the single-rail embodiment non-transferability is additionally enforced by the token standard. Expiration is enforced at validation time — the validator treats a credit whose expiration time has passed as invalid and commits an EXPIRED 230 event — rather than by any scheduled ledger operation, ledgers having no native clock-driven execution.

[0031] A credit may be surfaced to the consumer through platform wallet passes or in-application display, including a human-readable statement of the offer, its expiration, and its non-transferable, no-cash-value character.

Redemption (FIG. 3)

[0032] At issuance, a credit secret 114 is provisioned to the consumer device 110. At redemption, the application 112 derives a single-use code from the secret and the current time interval according to a time-based one-time-password algorithm (for example, of the type standardized in IETF RFC 6238), and displays it as an optically readable code that refreshes each interval. Because derivation is local, the consumer device requires no network connectivity at the moment of redemption; this is significant because points of settlement — interior retail locations — are frequently radio-shadowed.

[0033] The terminal 140 reads the presented code and the validator 144 verifies it against credential data. In connected operation the validator queries the settlement server 130, which verifies the code, verifies non-expiration, and executes an atomic commitment: in a single transaction, the credit's state is tested for BOUND and set to REDEEMED 220, and the bound instrument is extinguished (by burn or by anchored transition). The atomicity of the test-and-set guarantees that of any plurality of contemporaneous redemption attempts against one credit, exactly one succeeds. The terminal then presents a definitive indication and, where configured, actuates dispensing (paragraph [0037]).

[0034] In disconnected operation (FIG. 3, branch 330), the terminal validates the one-time code against locally cached credential data, honors the redemption locally, and appends to offline queue 142 a commitment signed with the terminal's key, comprising the credit identifier, code interval, and timestamp. Upon reconnection the queue is transmitted and each queued commitment is committed in order; where two terminals honored the same credit during a disconnection window, the first-committed prevails and the later is rejected and reported as an exception. The economic exposure of the disconnection window is bounded by the face value of individual promotional credits, and the design accepts this bounded exposure in exchange for the guarantee that the point of sale is never halted by network state.

[0035] The system provides no reversal: neither the settlement server 130 nor any ledger program exposes a transaction type that returns a REDEEMED or EXPIRED credit to BOUND, reinstates an extinguished instrument, or re-issues a credit under the same identifier. Because a promotional credit costs its holder nothing, no party requires reversal for protection; its absence eliminates the duplicate-collapse pathway and the dispute and clawback machinery of centralized clearing. Erroneous grants are remedied prospectively, by issuing a new credit, never by un-committing a prior event.

Settlement finality and issuance velocity

[0036] Immediate finality alters issuer economics. Under clearing-based settlement, every outstanding instrument carries weeks of indeterminate liability, and issuers throttle volume accordingly. In the disclosed system the issuer's knowledge of each instrument's terminal state is contemporaneous with the state itself;

an issuer may therefore replenish, retarget, and re-issue promptly upon observing consumption, operating promotional programs at a velocity that clearing-based settlement cannot support at equivalent risk.

Physical actuation (dispenser 150)

[0037] In embodiments deployed at amusement venues, commitment of a redemption causes terminal 140 to actuate dispenser 150 to dispense physical articles 152 — in one embodiment, metal amusement tokens of a custom diameter and alloy accepted only by the operator's machines, which recirculate through those machines and are re-dispensed. The extinguishment of the digital instrument and the release of the physical article are two halves of one settlement event; other physical actuations (sample release, receipt printing, gate release) are equivalent.

Event record and measurement (FIG. 4)

[0038] Every lifecycle transition is committed as an event to an append-only record 420: issuance 411, redemption 412, expiration 413, reimbursement 414, funding 415, attestation 416, impression digest 417, and fraud flag 418. Reimbursement and funding events permit multi-party settlement to be computed from the record itself: where a manufacturer funds an offer honored by an independent retailer, the retailer's reimbursement entitlement is computed automatically from committed redemption events, without a clearing intermediary and without either party trusting the other's private records.

[0039] Privacy is preserved structurally. Impression-type events, where used, are accumulated off-ledger and committed only as batched pseudonymous digests 417; identity verification is committed only as a boolean attestation 416; and no personally identifying information is written to any ledger. The record is a record of settlement facts concerning instruments, not of behavioral profiles concerning persons.

[0040] Because the record is complete rather than sampled, aggregation 430 yields population-scale measurement 440 of promotional activity in substantially real time: redemption velocity 441, breakage rate 442, and per-offer consumption 443, among others, computed over every instrument rather than estimated from panels. The measurement is a byproduct of settlement, requiring no additional data collection.

Consumer interface modes

[0041] In embodiments the application 112 operates in a first mode presenting credits, offers, and redemption without ledger terminology, and a second mode additionally exposing addresses, verification status, and lock-vault details, the modes differing in presentation only; the settlement path is identical in both.

Alternatives and scope

[0042] The disclosure is not limited to any particular ledger, token standard, code-presentation channel (optical, near-field, network), terminal hardware, or benefit class. Steps described as performed by the settlement server may be distributed among ledger programs (smart contracts) and server components in

any combination preserving the atomic test-and-set of paragraph [0033]. Recitation of specific commercial names, periods, quantities, and venues is by way of example. Numerous variations will be apparent to persons of ordinary skill, and the invention is defined by the claims rather than by the foregoing embodiments.

CLAIMS

What is claimed is:

1. A computer-implemented method of settling a promotional credit, comprising:

verifying, via a read-only query to a distributed ledger, that an account associated with a consumer holds at least a predetermined quantity of a membership asset, wherein the membership asset is not transferred or consumed by any step of the method;

issuing, in response to the verifying, a promotional credit to the consumer, the promotional credit being bound one-to-one to a unique settlement instrument, being non-transferable, and having an expiration time;

generating, on a consumer device and without requiring network connectivity of the consumer device at a time of redemption, a single-use redemption code cryptographically derived from a secret associated with the promotional credit;

validating the single-use redemption code at a redemption terminal at a point of settlement;

in response to the validating, irreversibly committing, by an atomic operation, a state transition of the unique settlement instrument that extinguishes the promotional credit, the committed state transition being verifiable by at least one party other than an issuer of the promotional credit; and

rejecting any subsequent attempt to redeem the promotional credit based on the committed state transition.

2. The method of claim 1, wherein the membership asset is held in a time-locked vault recorded on the distributed ledger, the vault preventing transfer of the membership asset for at least a predetermined period of years.

3. The method of claim 2, further comprising, prior to the verifying: receiving confirmation of a purchase of a digital item through a host application platform; and transferring, by an issuer treasury and in response to the confirmation, the predetermined quantity of the membership asset into the time-locked vault in association with the account.

4. The method of claim 1, wherein the unique settlement instrument comprises a non-transferable token recorded on the distributed ledger, and wherein committing the state transition comprises executing a burn transaction of the token.

5. The method of claim 1, wherein the promotional credit is recorded on a private ledger of a settlement system, and wherein committing the state transition comprises committing the transition on the private ledger and anchoring, to the distributed ledger, a cryptographic digest attesting to the transition, the digest attesting to a plurality of state transitions in a single anchoring operation.

6. The method of claim 1, wherein generating the single-use redemption code comprises deriving the code from the secret and a current time interval according to a time-based one-time-password algorithm, the code being valid for a single time interval.
7. The method of claim 1, wherein, during an interval in which the redemption terminal lacks network connectivity, the validating is performed against credential data cached at the redemption terminal, and further comprising: appending, to a store-and-forward queue of the redemption terminal, a commitment signed with a key of the redemption terminal; transmitting the queue upon restoration of connectivity; and, where a plurality of queued commitments concern the same promotional credit, committing only the earliest and rejecting each other commitment.
8. The method of claim 1, further comprising determining, at the validating, whether a current time exceeds the expiration time, and, when exceeded, committing an expiration event that extinguishes the promotional credit without honoring redemption.
9. The method of claim 1, wherein no component of the method exposes an operation that reverses a committed state transition, reinstates an extinguished settlement instrument, or re-issues a promotional credit under an identifier of an extinguished promotional credit.
10. A promotional settlement system comprising:
- a settlement server comprising a credit ledger and an offer engine;
 - a read-only interface to a distributed ledger, configured to determine whether an account holds at least a predetermined quantity of a membership asset without transferring or consuming the membership asset;
 - a consumer application executable on a consumer device and configured to generate, without network connectivity of the consumer device, single-use redemption codes derived from a locally stored secret; and
 - a redemption terminal configured to validate a presented single-use redemption code;
- wherein the system is configured to bind each issued promotional credit one-to-one to a unique settlement instrument, to irreversibly commit, by an atomic operation upon validation, a state transition extinguishing the credit, the transition being verifiable by at least one party other than an issuer, and to reject any subsequent redemption attempt against the credit.
11. The system of claim 10, further comprising a dispenser coupled to the redemption terminal and configured to dispense a physical article in response to commitment of the state transition.
12. The system of claim 11, wherein the physical article comprises a machine-usable token dimensioned and composed to be accepted only by amusement machines of an operator of the system, the token being recollected from the machines and re-dispensed.

- 13.** The system of claim 10, wherein the read-only interface is the only interface of the system to consumer state on the distributed ledger, the sole query of the interface being whether the account holding meets the predetermined quantity, and wherein the system writes no personally identifying information to the distributed ledger.
- 14.** The system of claim 10, wherein the consumer application is operable in a first mode that presents credits and redemption without distributed-ledger terminology and a second mode that additionally presents account addresses and verification detail, the settlement path being identical in both modes.
- 15.** The system of claim 10, wherein the settlement server is configured to commit each of a plurality of marketing event types to an append-only event record, the event types comprising at least issuance, redemption, and expiration of promotional credits, and further comprising at least one of a reimbursement event, a funding event, a verification-attestation event, and a fraud-flag event.
- 16.** The system of claim 15, wherein impression events are accumulated off-ledger and committed to the event record only as batched pseudonymous digests.
- 17.** The method of claim 1, further comprising deriving, from committed events across a population of accounts and in substantially real time, aggregate measurements of promotional activity comprising at least one of redemption velocity, breakage rate, and per-offer consumption, the deriving performed without access to personally identifying information.
- 18.** The method of claim 17, further comprising computing, from committed redemption events, a reimbursement obligation of a funding party to an honoring party, and committing a corresponding reimbursement event, whereby settlement between the funding party and the honoring party is determined from the committed events without a clearing intermediary.
- 19.** The method of claim 1, wherein verifying comprises committing a verification event consisting of a boolean attestation of eligibility, no personally identifying information being written to the distributed ledger.
- 20.** A non-transitory computer-readable medium storing instructions that, when executed by one or more processors of a settlement system, cause the settlement system to perform the method of claim 1.

ABSTRACT

A settlement system for promotional credits and related marketing events binds each credit one-to-one to a unique token recorded on a distributed ledger. Eligibility to receive credits is verified by confirming that a consumer account holds a separate membership asset that is never spent or transferred. Each credit is non-transferable, time-limited, and redeemable exactly once: at a point of settlement, a code generated on the consumer's device without network connectivity is validated by a redemption terminal, and the corresponding token state is irreversibly committed, extinguishing the credit. The system provides no reversal pathway, preventing duplicate redemption and counterfeiting without chargeback infrastructure. Terminals operate offline by queuing signed commitments for later synchronization. Committed events, including issuance, redemption, expiration, reimbursement, and funding events, form an auditable event-level record from which population-scale measurements of promotional activity are derived. Redemption may trigger dispensing of physical articles.

REFERENCES FOR COUNSEL'S PRIOR-ART SEARCH

Identified by the inventor by assignee and subject only; patent and publication numbers deliberately omitted pending professional search and verification. The distinguishing position for each family is stated for search calibration.

Balance-based distributed-ledger loyalty systems. Filings believed held by American Express (blockchain-based membership-rewards programs), International Business Machines Corp. (blockchain loyalty point allocation and exchange), Bank of America (cryptocurrency-related customer rewards), and Loyyal Corp. (blockchain loyalty interoperability). Distinguish: these record fungible point balances reconciled in aggregate; none binds an individual promise one-to-one to a unique instrument whose terminal state is individually and irreversibly committed at the point of settlement.

Centralized single-use digital coupon validation. Filings believed held by Quotient Technology / Coupons.com and by incumbent coupon-clearing providers (single-use codes, print-limit controls, POS validation). Distinguish: once-only enforcement resides solely in the issuer's private database; no commitment verifiable by non-issuer parties; reversal and dispute machinery retained; no offline-capable atomic settlement.

Retail digital-currency and payment-token systems. Filings believed held by Walmart (digital currency systems) and by numerous payment-network assignees. Distinguish: payment instruments presuppose a payer requiring recourse, and accordingly teach reversal, refund, and chargeback infrastructure; the present disclosure claims the deliberate absence of reversal pathways for an instrument class with no payer (claim 9).

Non-patent literature. Solana Token-2022 program extensions (non-transferable tokens); token-vesting and time-lock protocols on Solana (e.g., Streamflow); IETF RFC 6238 (TOTP) and RFC 4226 (HOTP); industry fraud and breakage analyses (Infosys BPM; Mastercard-affiliated promotion-abuse research; Gartner loyalty-breakage estimates) cited in the Background as motivation.

Search emphasis requested. (a) any art binding one promotional promise to one burnable ledger token with point-of-sale burn; (b) offline store-and-forward redemption against ledger-committed instruments; (c) never-consumed ledger-asset gating ("hold but never spend") for eligibility; (d) settlement systems affirmatively lacking reversal operations; (e) physical dispensing actuated by ledger commitment.